(Please write your Enrollment Number) Enrollment No. 16301172024

MID-TERM EXAMINATION

(Course Name: B. Tech. ECE/AI-ML/CSE-AI/ECE-AI) (Semester: I)

(October 2024) OFF LINE mode

Subject Code: BAS 103	Subject: Probability and Statistics
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	
	a) The odds that a book will be favorably reviewed by three independent critics are 5 to 2, 4 to 3 and 3 to 4 respectively. What is the probability that, of the three reviews, a majority will be favorable?		CO1
	b) Three bags A,B and C contains 4 red, 3 black, 2 white; 3 red ,4 black, 4 white; and 5 red , 2 black, 6 white balls respectively. If a bag is selected at random and a ball is drawn from it, find the probability that the ball drawn is red.		CO1
	c) Prove that the moment generating function of the sum of a number of independent random variables is equal to the product of their respective moment generating functions.		CO2
	d) Let X be the random variable with the following probability distribution:		CO2

X	-3	6	9
P(X)	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{3}$

Find $E(X)$ and $E(X^2)$and using the law of expectations, evaluate $E(2X+1)^2$.

	UNIT I		CO2
Q2	Attempt any two parts	(5*2=10)	
	a) Calculate the mean, median and mode for the following distribution:		

Monthly Wages in Rs.	No. of Workers
50-70	4
70-90	44
90-110	38
110-130	28
130-150	6
150-170	8

	170-190	12		
	190-210	2		
	210-230	2		

b) Calculate the first four moments of the following distributions about the mean and hence find $\beta_1 and\ \beta_2$.

x	0	1	2	3	4	5	6	7	8
f	1	8	28	56	70	56	28	8	1

c) A random variable X has the following probability function:

X	0	1	2	3	4	5	6	7
P(X)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

(i) Find k,

(ii) $P(3 < X \leq 6)$

(iii) Minimum value of x for which $P(X \leq x) \geq {}^1/_2$.

	UNIT II		CO2
Q3	Attempt any two parts	(5*2=10)	

a) If X and Y are two random variables having joint density function:

$$f(x,y) = \begin{cases} \frac{1}{8}(6-x-y), 0 \leq x < 2, 2 \leq y < 4 \\ 0, \quad Otherwise \end{cases}$$

Find $(i)P(X < 1 \cap Y < 3)(ii)P(X + Y < 3)\ (iii)P(X < 1|Y < 3)$

b) Two ideal dice are thrown. Let X_1 be the score on the first die and X_2 be the score on second die. Let Y denote the maximum of X_1 and X_2. Write the joint distribution of Y and X_1. Find the mean and variance of Y and $Cov(Y, X_1)$.

c) The first of the two samples has 100 items with mean 15 and standard deviation 3. If the whole group has 250 items with mean 15.6 and standard deviation $\sqrt{13.44}$. Find the standard deviation of the second group.